Black-Box Fast Multipole Method (bbFMM)

Detailed Notes

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Abstract

These notes summarize the Chebyshev-based low-rank interpolation technique used in the black-box FMM (bbFMM). The material follows Fong & Darve (2009) and is organized for project-report clarity.

Contents

1 Problem statement **3**
1.1 Computational challenge . . . 3

2 Chebyshev interpolation for kernel compression **3**
2.1 Chebyshev nodes . . . 3
2.2 Interpolation basis . . . 3
2.3 Low-rank kernel approximation (1D) . . . 4
2.4 Extension to d dimensions . . . 4

3 Hierarchy and spatial subdivision **4**
3.1 Near-field and far-field interactions . . . 5
3.2 Well-separatedness criterion . . . 5

4 Low-rank representation for a box pair **5**
4.1 M2L sampling matrix . . . 6
4.2 Multipole coefficients: P2M (Particle-to-Multipole) . . . 6
4.3 Local coefficients: M2L (Multipole-to-Local) . . . 6

5 Multilevel bbFMM algorithm **6**
5.1 Upward pass (P2M + M2M) . . . 7
5.1.1 P2M (Particle-to-Multipole) . . . 7
5.1.2 M2M (Multipole-to-Multipole) . . . 7
5.2 Interaction pass (M2L) . . . 8
5.3 Downward pass (L2L + L2P) . . . 9
5.3.1 L2L (Local-to-Local) . . . 9
5.3.2 L2P (Local-to-Particle) . . . 10

6 Worked 1D example **10**
6.1 Step 1: Anterpolation (P2M) 10
6.2 Step 2: Evaluation at Chebyshev nodes (M2L) 10
6.3 Step 3: Interpolation to evaluation points (L2P) 11

7 Key properties **11**
7.1 Computational complexity 11
7.2 Advantages of Chebyshev interpolation 12
7.3 Interaction lists 13

8 Algorithm summary **13**

1 Problem statement

Consider the problem of evaluating summations of the form

$$f(x_i) = \sum_{j=1}^{N} K(x_i, y_j)\, \sigma_j, \qquad i = 1, \ldots, N, \tag{1}$$

where:

- $x_i \in \mathbb{R}^d$ = evaluation points (locations where we compute the sum),
- $y_j \in \mathbb{R}^d$ = source points (locations that contribute to the sum),
- $\sigma_j \in \mathbb{R}$ = weights or strengths associated with each source,
- $K(x, y) : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ = interaction kernel (a smooth, non-oscillatory function),
- N = total number of points,
- d = spatial dimension (typically $d = 1, 2, 3$),
- $f(x_i)$ = resulting value at evaluation point x_i.

1.1 Computational challenge

Direct evaluation requires computing N sums, each with N terms, yielding a computational complexity of $\mathcal{O}(N^2)$. For large N, this becomes intractable. The bbFMM uses hierarchical spatial decomposition combined with Chebyshev interpolation to reduce this complexity to $\mathcal{O}(N)$ or $\mathcal{O}(N \log N)$.

2 Chebyshev interpolation for kernel compression

2.1 Chebyshev nodes

On the interval $[-1, 1]$, the n Chebyshev nodes are defined as the roots of the n-th Chebyshev polynomial $T_n(x)$:

$$x_m = \cos\left(\frac{(2m-1)\pi}{2n}\right), \qquad m = 1, \ldots, n. \tag{2}$$

Variable definitions:

- $T_n(x)$ = Chebyshev polynomial of the first kind, degree n
- x_m = m-th Chebyshev node in $[-1, 1]$
- n = number of interpolation nodes (accuracy parameter)

2.2 Interpolation basis

The Chebyshev interpolation basis function is defined as

$$S_n(x_l, x) = \frac{1}{n} + \frac{2}{n} \sum_{k=1}^{n-1} T_k(x_l)\, T_k(x), \tag{3}$$

where:

- $S_n(x_l, x)$ = interpolation kernel from node x_l to point x
- $T_k(x)$ = Chebyshev polynomial of degree k
- x_l = a Chebyshev node (the interpolation point)
- x = arbitrary point in $[-1, 1]$ where we evaluate

(This formula is the truncated Chebyshev reproducing kernel used for interpolation; see, e.g., Fong & Darve (2009), Eq. (13), or standard texts in approximation theory.)

2.3 Low-rank kernel approximation (1D)

For well-separated intervals, the kernel $K(x, y)$ can be approximated using interpolation in both arguments:

$$K(x, y) \approx \sum_{l=1}^{n} \sum_{m=1}^{n} K(x_l, y_m)\, S_n(x_l, x)\, S_n(y_m, y). \tag{4}$$

Variable definitions:

- x_l = l-th Chebyshev node in the evaluation interval I
- y_m = m-th Chebyshev node in the source interval J
- $K(x_l, y_m)$ = kernel sampled at the Chebyshev node pairs
- $S_n(x_l, x)$ = basis function evaluating contribution from node x_l to point x
- $S_n(y_m, y)$ = basis function evaluating contribution from source y to node y_m

Interpretation: This expression represents a *numerical low-rank approximation* of K. Instead of storing n^2 evaluations of K, we compute at n nodes in each dimension and interpolate.

2.4 Extension to d dimensions

In d dimensions, tensor products are used: each box has n^d Chebyshev nodes. The d-dimensional interpolation basis is

$$R_n(\mathbf{x}_l, \mathbf{x}) = \prod_{i=1}^{d} S_n(x_{l,i}, x_i), \tag{5}$$

where $\mathbf{x} = (x_1, \ldots, x_d)$ and $\mathbf{x}_l = (x_{l,1}, \ldots, x_{l,d})$ represents a tensor-product Chebyshev node.

3 Hierarchy and spatial subdivision

The computational domain is recursively subdivided into a tree structure of boxes (or cells):

- **Level 0:** Root box containing the entire domain
- **Level ℓ:** Leaf level (finest subdivision) with few points per box
- **Level k:** Intermediate levels, $0 < k < \ell$

Each box at level k is subdivided into 2^d children at level $k + 1$ (e.g., 8 children in 3D).

3.1 Near-field and far-field interactions

For computational efficiency, interactions are classified based on spatial separation:

Near-field (direct computation): Interactions between points in the same box or neighboring boxes. These are computed directly without approximation.

Far-field (approximated): Interactions between well-separated boxes, handled via the low-rank Chebyshev approximation.

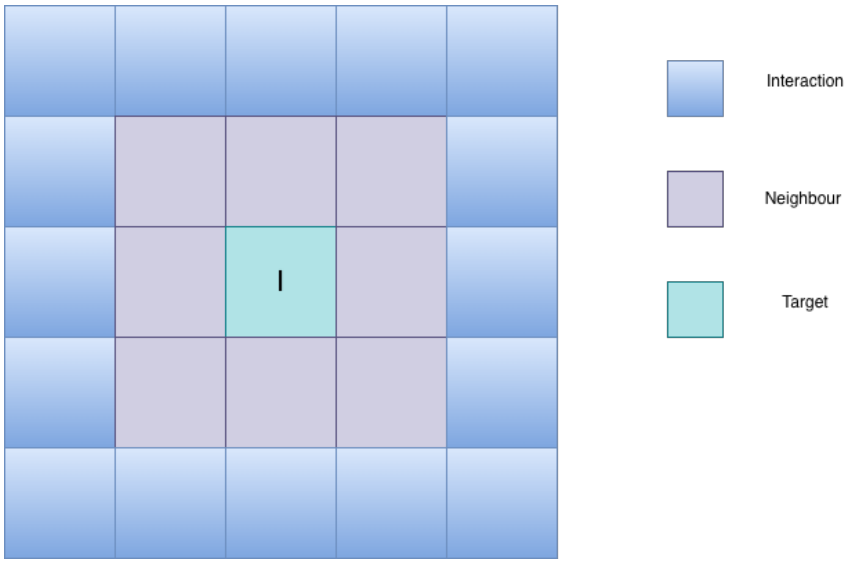

Figure 1: Classification of interactions into near-field (direct) and far-field (via M2L).

3.2 Well-separatedness criterion

Two boxes S and T are considered *well-separated* if

$$\text{dist}(S, T) \geq \varepsilon \cdot \min\{\text{diam}(S), \text{diam}(T)\}, \tag{6}$$

where:

- $\text{dist}(S, T)$ = minimum distance between any two points in boxes S and T
- $\text{diam}(S)$ = diameter of box S (maximum distance between any two points in S)
- $\varepsilon > 0$ = separation parameter (typically $\varepsilon \geq 1$)

Practical criterion: In the standard FMM tree, boxes are well-separated if they are non-overlapping and their parents are neighbors. This ensures the kernel K is sufficiently smooth over the interaction region for accurate interpolation.

4 Low-rank representation for a box pair

Consider a source box J and an evaluation box I, both at the same tree level, with their respective Chebyshev nodes:

- $\{x_l^{(I)}\}_{l=1}^{n^d}$ = Chebyshev nodes in evaluation box I
- $\{y_m^{(J)}\}_{m=1}^{n^d}$ = Chebyshev nodes in source box J

4.1 M2L sampling matrix

Define the *multipole-to-local sampling matrix*:

$$K_{lm}^{(I,J)} = K\Big(x_l^{(I)},\, y_m^{(J)}\Big), \qquad l, m = 1, \ldots, n^d, \tag{7}$$

where:

- $K_{lm}^{(I,J)}$ = kernel evaluated between node l in box I and node m in box J
- This is an $n^d \times n^d$ matrix for each pair (I, J)

4.2 Multipole coefficients: P2M (Particle-to-Multipole)

For a leaf box J, compute multipole coefficients at Chebyshev nodes by *anterpolation*:

$$W_m^{(J)} = \sum_{y_j \in J} \sigma_j\, S_n\big(y_m^{(J)}, y_j\big), \qquad m = 1, \ldots, n^d, \tag{8}$$

where:

- $W_m^{(J)}$ = multipole coefficient at node m of box J
- y_j = actual source points contained in box J
- σ_j = weight/strength of source j
- $S_n(y_m^{(J)}, y_j)$ = Chebyshev basis evaluating the contribution of source y_j to node $y_m^{(J)}$

Interpretation: This step aggregates information from all sources in box J onto the n^d Chebyshev nodes, creating a compressed representation.

4.3 Local coefficients: M2L (Multipole-to-Local)

For evaluation box I and source box J in the interaction list $\mathcal{I}(I)$:

$$g_l^{(I)} = \sum_{J \in \mathcal{I}(I)} \sum_{m=1}^{n^d} K_{lm}^{(I,J)}\, W_m^{(J)}, \qquad l = 1, \ldots, n^d, \tag{9}$$

where:

- $g_l^{(I)}$ = local expansion coefficient at node l of box I
- $\mathcal{I}(I)$ = interaction list (far-field boxes that interact with I)
- $K_{lm}^{(I,J)}$ = kernel samples between Chebyshev nodes
- $W_m^{(J)}$ = multipole coefficients from source box J

Interpretation: This computes the far-field contribution to box I from all well-separated source boxes, storing the result at Chebyshev nodes.

5 Multilevel bbFMM algorithm

The algorithm consists of three passes through the tree: upward, interaction, and downward.

5.1 Upward pass (P2M + M2M)

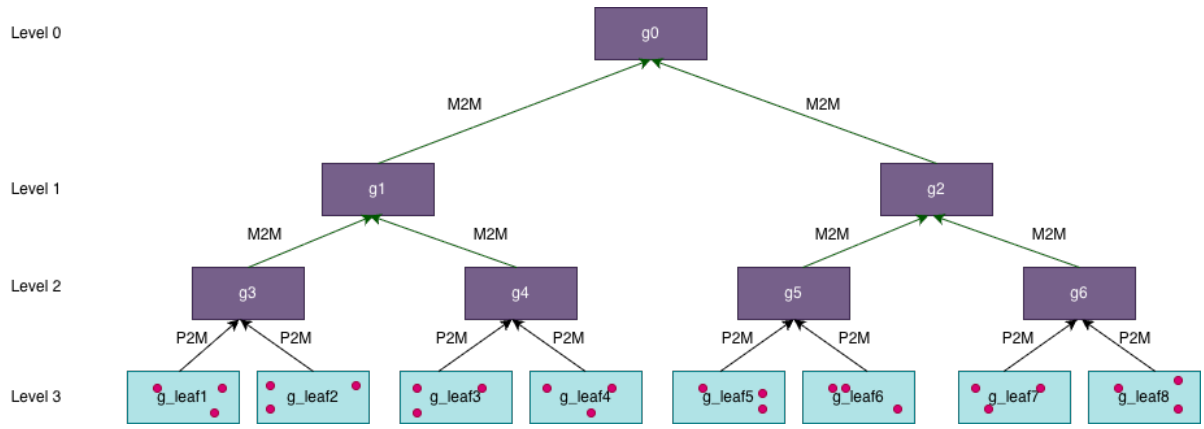

Figure 2: Upward pass: P2M aggregates sources to leaf nodes; M2M propagates multipole expansions up the tree.

5.1.1 P2M (Particle-to-Multipole)

For leaf boxes J at level ℓ:

$$W_m^{(J)} = \sum_{y_j \in J} \sigma_j \, S_n\big(y_m^{(J)}, \, y_j\big), \qquad m = 1, \ldots, n^d. \tag{10}$$

Cost: $\mathcal{O}(n^d)$ per source point, $\mathcal{O}(n^d N)$ total.

5.1.2 M2M (Multipole-to-Multipole)

For each parent box P with children c, starting from level $\ell - 1$ down to level 0:

$$W_m^{(P)} = \sum_{c \in \text{children of } P} \sum_{m'=1}^{n^d} A_{m,m'}^{(P,c)} \, W_{m'}^{(c)}, \tag{11}$$

with the translation operator

$$A_{m,m'}^{(P,c)} = S_n\big(y_m^{(P)}, y_{m'}^{(c)}\big), \tag{12}$$

where:

- $A_{m,m'}^{(P,c)}$ = translation matrix from child c to parent P
- $y_m^{(P)}$ = m-th Chebyshev node in parent box P
- $y_{m'}^{(c)}$ = m'-th Chebyshev node in child box c
- $W_{m'}^{(c)}$ = multipole coefficient from child
- $W_m^{(P)}$ = resulting multipole coefficient at parent

Interpretation: This shifts the compressed representation from finer to coarser levels, maintaining the multipole expansion at each level.

Cost: $\mathcal{O}(n^{2d})$ per box, $\mathcal{O}(n^{2d} N)$ total across all levels.

5.2 Interaction pass (M2L)

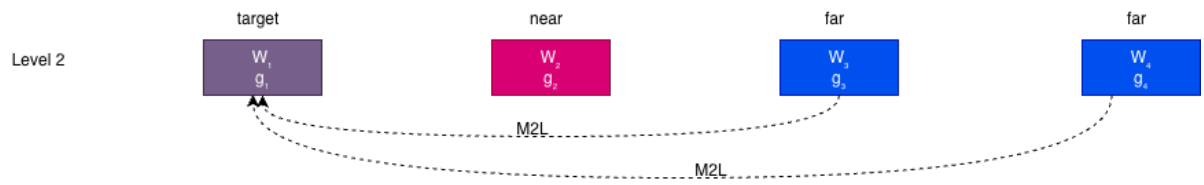

Figure 3: Interaction pass: M2L operations convert far-field multipole expansions to local expansions.

For each box I at level k, $0 \leq k \leq \ell$:

$$g_l^{(I)} = \sum_{J \in \mathcal{I}(I)} \sum_{m=1}^{n^d} K_{lm}^{(I,J)} \, W_m^{(J)}, \qquad l = 1, \ldots, n^d, \tag{13}$$

where:

- $\mathcal{I}(I)$ = far-field interaction list of box I
  - In 3D: well-separated boxes at the same level whose parents are neighbors of I's parent
  - Up to 189 boxes in 3D (standard bbFMM constant; see Fong & Darve, 2009)
- $g_l^{(I)}$ = local expansion coefficient at Chebyshev node l of box I
- $K_{lm}^{(I,J)}$ = kernel sampled between node pairs
- $W_m^{(J)}$ = multipole coefficients from source box J

Cost: $\mathcal{O}(n^{2d})$ per interaction, $\mathcal{O}(n^{2d}N)$ total (dominant operation).

5.3 Downward pass (L2L + L2P)

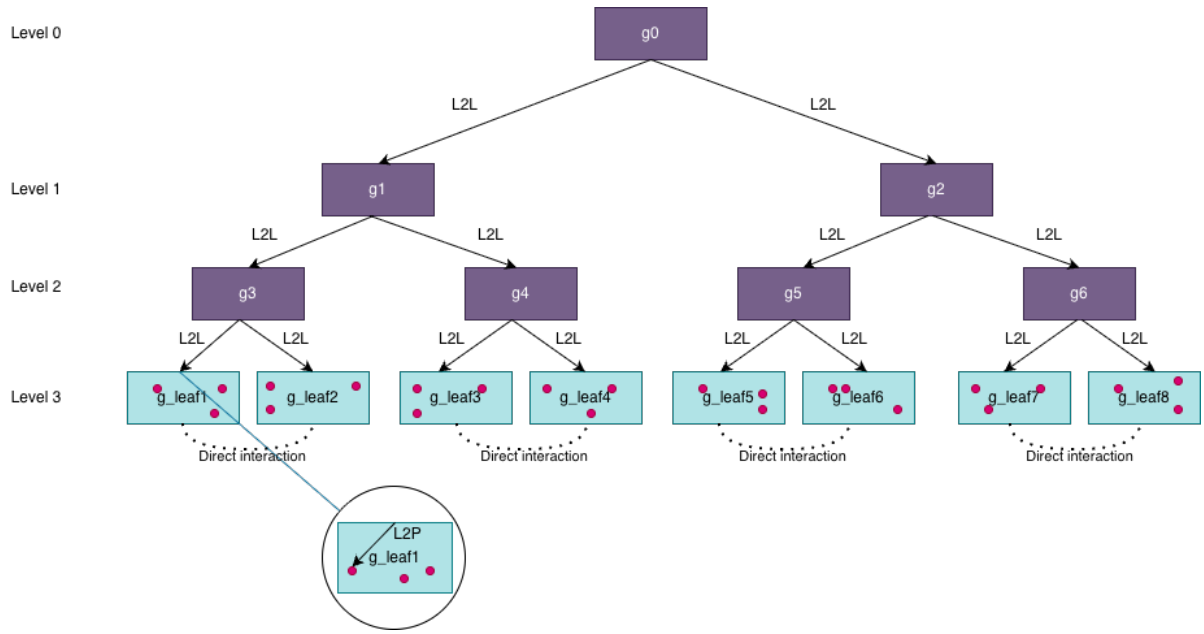

Figure 4: Downward pass: L2L propagates local expansions down the tree; L2P evaluates the potential at particle locations in leaf boxes.

5.3.1 L2L (Local-to-Local)

For child box c with parent P, starting from level 1 down to level ℓ:

$$g_l^{(c)} = g_l^{(c)} + \sum_{l'=1}^{n^d} B_{l,l'}^{(c,P)}\, g_{l'}^{(P)}, \qquad l = 1, \ldots, n^d, \tag{14}$$

with the translation operator

$$B_{l,l'}^{(c,P)} = S_n\big(x_l^{(c)}, x_{l'}^{(P)}\big), \tag{15}$$

where:

- $B_{l,l'}^{(c,P)}$ = translation matrix from parent P to child c
- $x_l^{(c)}$ = l-th Chebyshev node in child box c
- $x_{l'}^{(P)}$ = l'-th Chebyshev node in parent box P
- $g_{l'}^{(P)}$ = local expansion coefficient from parent
- $g_l^{(c)}$ = accumulated local expansion coefficient at child (updated)

Interpretation: This refines the far-field approximation by translating from coarse to fine levels, accumulating contributions.

Cost: $\mathcal{O}(n^{2d})$ per box, $\mathcal{O}(n^{2d}N)$ total.

5.3.2 L2P (Local-to-Particle)

For evaluation points x_i in leaf box I:

$$f(x_i) = \underbrace{\sum_{l=1}^{n^d} g_l^{(I)}\, S_n(x_l^{(I)}, x_i)}_{\text{far-field}} + \underbrace{\sum_{J \in \mathcal{N}(I)} \sum_{y_j \in J} \sigma_j\, K(x_i, y_j)}_{\text{near-field}}, \tag{16}$$

where:

- First term: far-field contribution via local expansion interpolation
- $g_l^{(I)}$ = local expansion coefficients at Chebyshev nodes
- $S_n(x_l^{(I)}, x_i)$ = Chebyshev basis interpolating from node to target
- Second term: near-field direct evaluation
- $\mathcal{N}(I)$ = near-field neighbor boxes (including I itself, typically 27 boxes in 3D)
- Direct computation for nearby interactions where interpolation is inaccurate

  Cost: $\mathcal{O}(n^d)$ per evaluation point (far-field) + $\mathcal{O}(1)$ per near-field pair, $\mathcal{O}(N)$ total.

6 Worked 1D example

Consider a simple 1D scenario with source interval $J = [0, 1]$ and evaluation interval $I = [2, 3]$, using $n = 3$ Chebyshev nodes.

6.1 Step 1: Anterpolation (P2M)

Chebyshev nodes in $J = [0, 1]$ (mapped from $[-1, 1]$):

$$y_1 = 0.5 - 0.5\cos\left(\frac{\pi}{6}\right) \approx 0.067$$
$$y_2 = 0.5 - 0.5\cos\left(\frac{\pi}{2}\right) = 0.5$$
$$y_3 = 0.5 - 0.5\cos\left(\frac{5\pi}{6}\right) \approx 0.933$$

Suppose sources $\{y_j\}$ are located at $\{0.1, 0.4, 0.8\}$ with weights $\{\sigma_j\} = \{1, -2, 1\}$. Compute multipole coefficients:

$$W_m = \sum_{j=1}^{3} \sigma_j\, S_3(y_m, y_j), \qquad m = 1, 2, 3, \tag{17}$$

where S_3 evaluates the Chebyshev interpolation basis.

6.2 Step 2: Evaluation at Chebyshev nodes (M2L)

Chebyshev nodes in $I = [2, 3]$:

$$x_1 \approx 2.067, \quad x_2 = 2.5, \quad x_3 \approx 2.933$$

Compute local coefficients using kernel $K(x, y) = \frac{1}{|x-y|}$:

$$g_l = \sum_{m=1}^{3} K(x_l, y_m)\, W_m, \qquad l = 1, 2, 3. \tag{18}$$

6.3 Step 3: Interpolation to evaluation points (L2P)

For arbitrary evaluation point $x \in I$:

$$f(x) \approx \sum_{l=1}^{3} g_l \, S_3(x_l, x). \tag{19}$$

This example illustrates the three-stage process: compress sources to nodes (P2M), compute at nodes (M2L), and interpolate to targets (L2P).

Numerical values for the 1D example

Using the $n = 3$ Chebyshev nodes in $J = [0, 1]$:

$$y_1 \approx 0.0670, \qquad y_2 = 0.5, \qquad y_3 \approx 0.933.$$

The sources are located at $\{0.1, 0.4, 0.8\}$ with strengths $\{1, -2, 1\}$.
Evaluating the basis functions $S_3(y_m, y_j)$ and forming

$$W_m = \sum_{j=1}^{3} \sigma_j \, S_3(y_m, y_j),$$

gives the multipole coefficients:

$$W_1 \approx -0.158, \qquad W_2 \approx -0.842, \qquad W_3 \approx -0.158.$$

For the evaluation interval $I = [2, 3]$, the Chebyshev nodes are:

$$x_1 \approx 2.067, \qquad x_2 = 2.5, \qquad x_3 \approx 2.933.$$

Using the kernel $K(x, y) = 1/|x - y|$, the local coefficients

$$g_l = \sum_{m=1}^{3} K(x_l, y_m) \, W_m$$

are:

$$g_1 \approx -1.043, \qquad g_2 \approx -0.800, \qquad g_3 \approx -0.648.$$

Thus, for any target $x \in I$,

$$f(x) \approx g_1 \, S_3(x_1, x) + g_2 \, S_3(x_2, x) + g_3 \, S_3(x_3, x),$$

which provides the approximate potential obtained from the bbFMM interpolation pipeline.

7 Key properties

7.1 Computational complexity

We analyze the computational cost of the black-box FMM for a uniform distribution of N particles in d dimensions. The adaptive tree has depth $\ell = \mathcal{O}(\log N)$ and each finest-level box contains $\mathcal{O}(1)$ particles.

Each box stores n^d Chebyshev interpolation nodes. Since n depends only on the desired accuracy and not on N, it is treated as a small constant in the $N \to \infty$ regime.

A key quantity is the number of well-separated interaction partners per box, denoted I_d. Under the standard interaction-list definition for black-box FMMs:

$$I_1 = 2, \qquad I_2 = 27, \qquad I_3 = 189.$$

These are fixed dimension-dependent constants that arise from the parent–neighbor construction of the interaction list.

Operation	**Cost per box**	**Total cost**
P2M	$\mathcal{O}(n^d)$	$\mathcal{O}(n^d N)$
M2M	$\mathcal{O}(n^{2d})$	$\mathcal{O}(n^{2d} N)$
M2L	$\mathcal{O}(I_d\, n^{2d})$	$\mathcal{O}(I_d\, n^{2d} N)$
L2L	$\mathcal{O}(n^{2d})$	$\mathcal{O}(n^{2d} N)$
L2P	$\mathcal{O}(n^d)$	$\mathcal{O}(n^d N)$
Near-field (direct)	$\mathcal{O}(1)$ per particle	$\mathcal{O}(N)$
Overall		$\boldsymbol{\mathcal{O}(n^{2d} N)}$

Since n is independent of N, the algorithm achieves

$$\boxed{\mathcal{O}(N)} \quad \text{complexity.}$$

Dependence on accuracy. For kernels that are analytic on each pair of well-separated boxes, Chebyshev interpolation exhibits exponential convergence. The required interpolation order typically satisfies

$$n = \mathcal{O}\big(\log(1/\varepsilon)\big),$$

where ε is the desired accuracy. Substituting this into the dominant $n^{2d}N$ term yields

$$\boxed{\mathcal{O}\left(N \log^{2d}\frac{1}{\varepsilon}\right)}$$

as the accuracy-dependent complexity. For example, in three dimensions ($d = 3$),

$$\mathcal{O}\left(N \log^{6}\frac{1}{\varepsilon}\right).$$

Thus, for fixed accuracy the bbFMM scales linearly with N, while the accuracy dependence is polylogarithmic, making the method highly efficient for large-scale simulations.

7.2 Advantages of Chebyshev interpolation

1. **Black-box nature:** Only requires kernel evaluations $K(x, y)$; no analytical expansions needed
2. **Spectral accuracy:** Exponential convergence for smooth kernels
3. **Uniform error:** Near-minimax approximation across the entire box
4. **Stability:** Avoids Runge phenomenon (unlike equispaced nodes)
5. **Generality:** Same algorithm applies to all smooth, non-oscillatory kernels

7.3 Interaction lists

For a box I at level k:

- **Near-field list $\mathcal{N}(I)$:**
  - Box I and its immediate neighbors
  - In 3D: up to 27 boxes (3^3)
  - Evaluated via direct (particle–particle or node–node) computation
- **Far-field list $\mathcal{I}(I)$:**
  - Boxes at the same level that are not neighbors of I, but whose parents are neighbors of I's parent
  - In 3D: up to 189 boxes (standard bbFMM constant; see Fong & Darve, 2009)
  - Evaluated via M2L using Chebyshev interpolation

Note : Because kernels such as $K(x, y) = 1/|x - y|$ are singular on the diagonal, the Chebyshev interpolation used in the M2L step is accurate only for well-separated boxes; when boxes are adjacent the kernel loses smoothness, and near-field contributions must therefore be computed directly.

8 Algorithm summary

1. **Tree construction:**
   - Recursively subdivide domain into octree (3D) or quadtree (2D)
   - Continue until each leaf box contains $\leq s$ points (typically $s \approx$ 100-500)
   - Assign Chebyshev nodes to each box at all levels
2. **Upward pass:**
   - **P2M:** Compute multipole expansions at leaf boxes via anterpolation
   - **M2M:** Translate multipole expansions from children to parents up the tree
3. **Interaction pass:**
   - **M2L:** Compute far-field contributions via multipole-to-local translations
   - Process interaction list $\mathcal{I}(I)$ for each box I
4. **Downward pass:**
   - **L2L:** Translate local expansions from parents to children down the tree
   - **L2P:** Evaluate local expansion at target points via interpolation
5. **Near-field:**
   - Compute direct interactions for nearby particles in $\mathcal{N}(I)$
   - Add to far-field contribution from L2P

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